

PART A — QUESTIONS

AF18-001

Question:

Solve: $20x + 11 = 131$

- A) 5
- B) 6
- C) 7
- D) 8

AF18-002

Question:

If $18n - 9 = 117$, what is n ?

- A) 5
- B) 6
- C) 7
- D) 8

AF18-003

Question:

Simplify: $5(6x - 4) - 9x$

- A) $21x - 20$
- B) $30x - 20$
- C) $21x - 4$
- D) $39x - 20$

AF18-004

Question:

A company charges \$34 per service plus a \$28 setup fee. Which expression represents the total charge for s services?

- A) $34s + 28$
- B) $28s + 34$
- C) $62s$
- D) $34 - 28s$

AF18-005

Question:

If $f(x) = 15x - 37$, what is $f(8)$?

- A) 73
- B) 83
- C) 120
- D) 157

AF18-006

Question:

Which ordered pair satisfies the equation $y = 13x - 4$?

- A) (1, 8)
- B) (2, 22)
- C) (3, 35)
- D) (4, 50)

AF18-007

Question:

Solve: $20(x - 6) = 140$

- A) 11
- B) 12
- C) 13
- D) 14

AF18-008

Question:

A pattern begins: 6, 23, 40, 57, ... What is the next number?

- A) 72
- B) 73
- C) 74
- D) 75

AF18-009

Question:

Solve: $x/32 = 3$

- A) 35
- B) 64
- C) 96
- D) 128

AF18-010

Question:

A line has a slope of -15 and crosses the y-axis at 12. Which equation represents the line?

- A) $y = 12x - 15$
- B) $y = -15x + 12$
- C) $y = 15x + 12$
- D) $y = -12x + 15$

AF18-011

Question:

If 9 cases contain 261 parts, how many parts are in 7 cases at the same rate?

- A) 189
- B) 196
- C) 203
- D) 210

AF18-012

Question:

Solve: $210 - 24x = 114$

- A) 3
- B) 4
- C) 5
- D) 6

AF18-013

Question:

Which expression is equivalent to $80q - 37q - 11$?

- A) $43q - 11$
- B) $117q - 11$
- C) $43q + 11$
- D) $80q - 48$

AF18-014

Question:

A storage room starts with 1,050 units. Workers remove 70 units each day. Which equation gives the number of units U left after d days?

- A) $U = 1,050d - 70$
- B) $U = 70d + 1,050$
- C) $U = 1,050 - 70d$
- D) $U = 70d - 1,050$

AF18-015

Question:

If $y = -18x + 225$, what is y when $x = 11$?

- A) 17
- B) 27
- C) 198
- D) 423

AF18-016

Question:

Solve: $30x - 42 = 19x + 68$

- A) 8
- B) 9
- C) 10
- D) 11

AF18-017

Question:

Which value of x makes $16x + 5$ less than 165?

- A) 9
- B) 10
- C) 11
- D) 12

AF18-018

Question:

A table shows a linear relationship.

x : 0, 1, 2, 3

y : 6, 24, 42, 60

What is the rule?

- A) $y = 6x + 18$
- B) $y = 18x + 6$
- C) $y = 24x - 18$
- D) $y = x + 6$

AF18-019

Question:

Solve: $15(x + 6) = 12x + 126$

- A) 10
- B) 11
- C) 12
- D) 13

AF18-020

Question:

If $a = -11$ and $b = 10$, what is $4a + 5b$?

- A) -94
- B) -6
- C) 6
- D) 94

AF18-021

Question:

Which point is located on the x-axis?

- A) (0, -14)
- B) (-14, 0)
- C) (-14, 6)
- D) (14, -6)

AF18-022

Question:

A worker sorts 195 pieces in 13 minutes. At the same rate, how many pieces can the worker sort in 6 minutes?

- A) 75
- B) 84
- C) 90
- D) 96

AF18-023

Question:

Solve for m: $C = 25m + 75$

- A) $m = C - 75$
- B) $m = (C - 75)/25$
- C) $m = 25(C - 75)$
- D) $m = C/25 + 75$

AF18-024

Question:

What is the slope of a line that goes through (6, 10) and (12, 76)?

- A) 9
- B) 10
- C) 11
- D) 66

AF18-025

Question:

Simplify: $60x + 17 - 49x - 29$

- A) $11x - 12$
- B) $109x - 12$
- C) $11x + 46$
- D) $109x + 46$

AF18-026

Question:

If $g(x) = x^2 + 6x$, what is $g(11)$?

- A) 121
- B) 143
- C) 187
- D) 242

AF18-027

Question:

A sequence follows the rule “subtract 8, then multiply by 5.” If the first term is 16, what is the second term?

- A) 8
- B) 24
- C) 40
- D) 80

AF18-028

Question:

Solve: $20(x - 4) + 12 = 172$

- A) 10
- B) 11
- C) 12
- D) 13

AF18-029

Question:

A line crosses the y-axis at -17 and rises 8 units for every 1 unit to the right. Which equation represents the line?

- A) $y = -17x + 8$
- B) $y = 8x - 17$
- C) $y = -8x - 17$
- D) $y = 17x + 8$

AF18-030

Question:

If $24/30 = x/90$, what is x ?

- A) 60
- B) 66
- C) 72
- D) 78

AF18-031

Question:

Solve: $-25x + 100 = -175$

- A) -11
- B) -7
- C) 7
- D) 11

AF18-032

Question:

A warehouse starts with 3,600 washers. Each kit uses 180 washers. Which expression gives the number of washers left after k kits?

- A) $3,600 + 180k$
- B) $3,600 - 180k$
- C) $180k - 3,600$
- D) $3,600 \div 180k$

AF18-033

Question:

Which equation has $x = 24$ as its solution?

- A) $2x + 19 = 65$
- B) $5x - 30 = 90$
- C) $x/12 + 9 = 12$
- D) $3x - 15 = 60$

PART B — ANSWER KEY + EXPLANATIONS

AF18-001 — Correct: B — Explanation:

Solve $20x + 11 = 131$. Subtract 11 from both sides: $20x = 120$. Divide by 20: $x = 6$. The most common mistake is subtracting 11 correctly but forgetting to divide by 20.

AF18-002 — Correct: C — Explanation:

Start with $18n - 9 = 117$. Add 9 to both sides: $18n = 126$. Divide by 18: $n = 7$. A common mistake is subtracting 9 instead of adding it.

AF18-003 — Correct: A — Explanation:

Distribute first: $5(6x - 4) = 30x - 20$. Then subtract $9x$: $30x - 9x - 20 = 21x - 20$. A common mistake is forgetting to multiply -4 by 5.

AF18-004 — Correct: A — Explanation:

The company charges \$34 for each service, so that part is $34s$. The \$28 setup fee is added one time. The total charge is $34s + 28$. A common mistake is treating the setup fee as the per-service charge.

AF18-005 — Correct: B — Explanation:

Substitute 8 for x : $f(8) = 15(8) - 37 = 120 - 37 = 83$. A common mistake is adding 37 instead of subtracting it.

AF18-006 — Correct: C — Explanation:

Test the choices in $y = 13x - 4$. For $(3, 35)$, $13(3) - 4 = 39 - 4 = 35$, which matches the y -value. A common mistake is choosing a point that is close but does not make the equation true.

AF18-007 — Correct: C — Explanation:

Solve $20(x - 6) = 140$. Divide both sides by 20: $x - 6 = 7$. Add 6: $x = 13$. A common mistake is adding 6 before undoing the multiplication by 20.

AF18-008 — Correct: C — Explanation:

The pattern increases by 17 each time: 6, 23, 40, 57. Add 17 to 57: 74. A common mistake is assuming the increase is 16 because the numbers are close together.

AF18-009 — Correct: C — Explanation:

$x/32 = 3$ means x divided by 32 equals 3. Multiply both sides by 32: $x = 96$. A common mistake is adding 32 and 3 instead of multiplying.

AF18-010 — Correct: B — Explanation:

Slope-intercept form is $y = mx + b$. The slope is -15 and the y-intercept is 12, so the equation is $y = -15x + 12$. A common mistake is switching the slope and y-intercept.

AF18-011 — Correct: C — Explanation:

Find the unit rate: $261 \text{ parts} \div 9 \text{ cases} = 29 \text{ parts per case}$. For 7 cases: $29 \times 7 = 203$. A common mistake is multiplying 261 by 7 without finding the rate first.

AF18-012 — Correct: B — Explanation:

Solve $210 - 24x = 114$. Subtract 210 from both sides: $-24x = -96$. Divide by -24: $x = 4$. A common mistake is losing the negative sign after subtracting 210.

AF18-013 — Correct: A — Explanation:

Combine like terms: $80q - 37q = 43q$. The constant -11 stays the same. The expression becomes $43q - 11$. A common mistake is combining the constant with the variable terms.

AF18-014 — Correct: C — Explanation:

The storage room starts with 1,050 units. Workers remove 70 units each day, so after d days, 70d units are removed. The number left is $U = 1,050 - 70d$. A common mistake is adding 70d even though units are being removed.

AF18-015 — Correct: B — Explanation:

Substitute $x = 11$ into $y = -18x + 225$: $y = -18(11) + 225 = -198 + 225 = 27$. A common mistake is treating -18(11) as positive 198.

AF18-016 — Correct: C — Explanation:

Solve $30x - 42 = 19x + 68$. Subtract 19x: $11x - 42 = 68$. Add 42: $11x = 110$. Divide by 11: $x = 10$. A common mistake is adding 42 to only one side.

AF18-017 — Correct: A — Explanation:

Solve $16x + 5 < 165$. Subtract 5: $16x < 160$. Divide by 16: $x < 10$. The only listed value less than 10 is 9. A common mistake is choosing 10, but $16(10) + 5 = 165$, not less than 165.

AF18-018 — Correct: B — Explanation:

The y-values increase by 18 each time x increases by 1, so the slope is 18. When $x = 0$, $y = 6$, so the y-intercept is 6. The rule is $y = 18x + 6$. A common mistake is using 6 as the slope because it is the first y-value.

AF18-019 — Correct: C — Explanation:

Distribute: $15x + 90 = 12x + 126$. Subtract 12x: $3x + 90 = 126$. Subtract 90: $3x = 36$. Divide by 3: $x = 12$. A common mistake is forgetting to multiply 6 by 15.

AF18-020 — Correct: C — Explanation:

Substitute $a = -11$ and $b = 10$: $4a + 5b = 4(-11) + 5(10) = -44 + 50 = 6$. A common mistake is treating $4(-11)$ as positive 44.

AF18-021 — Correct: B — Explanation:

A point on the x-axis has $y = 0$. The point $(-14, 0)$ is on the x-axis. A common mistake is choosing $(0, -14)$, which is on the y-axis.

AF18-022 — Correct: C — Explanation:

195 pieces in 13 minutes means $195 \div 13 = 15$ pieces per minute. In 6 minutes: $15 \times 6 = 90$. A common mistake is multiplying 195 by 6 without finding the rate first.

AF18-023 — Correct: B — Explanation:

Start with $C = 25m + 75$. Subtract 75 from both sides: $C - 75 = 25m$. Divide by 25: $m = (C - 75)/25$. A common mistake is dividing C by 25 first and then subtracting 75.

AF18-024 — Correct: C — Explanation:

Slope equals change in y divided by change in x. From $(6, 10)$ to $(12, 76)$, y increases by 66 and x increases by 6. The slope is $66 \div 6 = 11$. A common mistake is using 66 as the slope without dividing by the change in x.

AF18-025 — Correct: A — Explanation:

Combine like terms: $60x - 49x = 11x$, and $17 - 29 = -12$. The simplified expression is $11x - 12$. A common mistake is writing $11x + 12$ because the negative sign before 29 is missed.

AF18-026 — Correct: C — Explanation:

Substitute 11 for x: $g(11) = 11^2 + 6(11) = 121 + 66 = 187$. A common mistake is calculating 11^2 as 22 instead of 121.

AF18-027 — Correct: C — Explanation:

Start with 16. Subtract 8: 8. Then multiply by 5: 40. A common mistake is multiplying first and then subtracting 8.

AF18-028 — Correct: C — Explanation:

Solve $20(x - 4) + 12 = 172$. Distribute: $20x - 80 + 12 = 172$. Combine: $20x - 68 = 172$. Add 68: $20x = 240$. Divide by 20: $x = 12$. A common mistake is forgetting to combine -80 and $+12$.

AF18-029 — Correct: B — Explanation:

The line rises 8 units for every 1 unit to the right, so the slope is 8. It crosses the y-axis at -17 , so the equation is $y = 8x - 17$. A common mistake is switching the slope and intercept.

AF18-030 — Correct: C — Explanation:

Use the proportion $24/30 = x/90$. Since $30 \times 3 = 90$, multiply 24 by 3: $x = 72$. A common mistake is adding 60 to the numerator because 30 becomes 90.

AF18-031 — Correct: D — Explanation:

Solve $-25x + 100 = -175$. Subtract 100 from both sides: $-25x = -275$. Divide by -25 : $x = 11$. A common mistake is giving -11 because both sides contain negative numbers.

AF18-032 — Correct: B — Explanation:

The warehouse starts with 3,600 washers. Each kit uses 180 washers, so k kits use $180k$ washers.

The number left is $3,600 - 180k$. A common mistake is adding $180k$ even though washers are being used.

AF18-033 — Correct: B — Explanation:

Test $x = 24$. In choice B, $5(24) - 30 = 120 - 30 = 90$, so the equation is true. A common mistake is choosing a nearby equation without checking both sides exactly.